LABORATORY DIAGNOSIS FOR ENTERIC BACTERIAL INFECTION

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OBJECTIVES

- Steps of stool culture for bacterial pathogens
- Characteristics of nontyphoidal *Salmonella*, *Shigella* spp. and *Vibrio* spp., *Campylobacter jejuni*, *Bacillus cereus, and Clostridioides/Clostridium difficile*
- Test of *Clostridioides/Clostridium difficile* toxins in stool



INDICATION OF STOOL CULTURE

- Moderate to severe diarrhea
  - eg. reduced activity or disability due to illness, hypovolemic shock, sepsis
- Dysentery
- Profuse cholera-like diarrhea
- Prolonged or persistent diarrhea (>14 d)
- Immunocompromised patients
- Outbreak



DIAGNOSTIC TESTS FOR ENTERIC BACTERIAL PATHOGENS

- **Stool culture**
  - Differential and selective: pathogen vs normal flora
- Serology
  - eg. *V. cholerae* O1 and O139; *Shigella* spp.; *E. coli* O157:H7
- Toxin detection
  - Toxin A/B for *C. difficile*
- molecular diagnosis eg. PCR, real-time PCR, microarray
  - eg. *E. coli* (eg. STEC/EHEC, ETEC, EPEC, EAEC, EIEC), *Salmonella, Shigella*
- Extraintestinal infection: other samples e.g. blood, CSF, etc.



STOOL COLLECTION



Don't let the sample go into the toilet



Collect the stool in some sort of clean container



Put a portion of the stool sample in the container given to you by the doctor

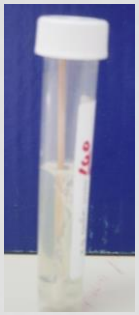


Wash hands well

Pictures from www.poopreport.com

- Fresh stool: sterile container
  - Send to the laboratory immediately
  - Store in a fridge (but for **no longer than 24 h**)
- Stool or rectal swab: transport medium
  - e.g. Cary-Blair transport medium
  - Room temperature





Laboratory activities

1. Stool culture

1.1 differential and selective agar

- MacConkey agar: *Salmonella* spp.
- SS agar: *Salmonella* spp., *Shigella* spp.
- MSRV agar: *Salmonella* spp.
- TCBS agar: *Vibrio* spp.

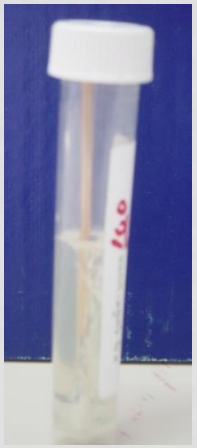
1.2 Enrichment broth

- tetrathionate broth
- alkaline peptone water

Stool Culture

Fresh stool





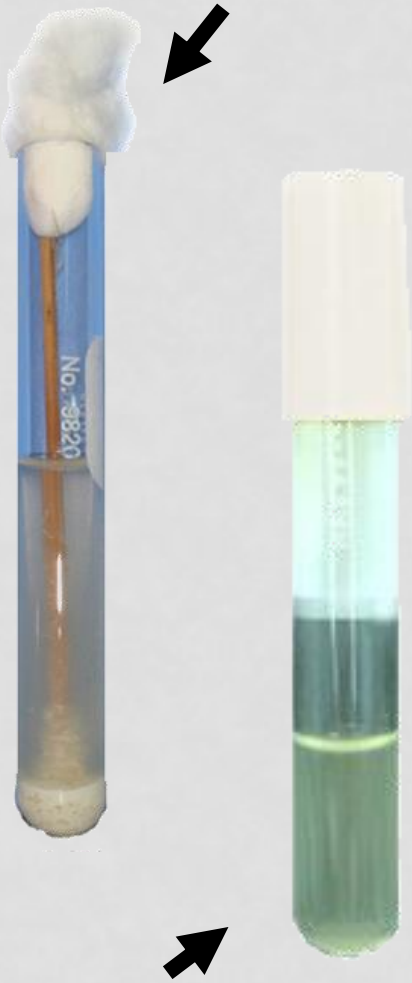
Cary-Blair transport medium

Differential/ Selective agar



Enrichment broth

Tetrathionate broth



Alkaline peptone water
37°C 24 hr

DIFFERENTIAL OR SELECTIVE MEDIUM

Differential medium :

- **Low selective media**: **MacConkey agar**
  - Bile salt (low concentration) and crystal violet : inhibit gram-positive bacteria and some fastidious gram-negative bacteria
  - Neutral red : indicator
  - **Lactose: CHO source**
  - **Gram-negative bacteria**

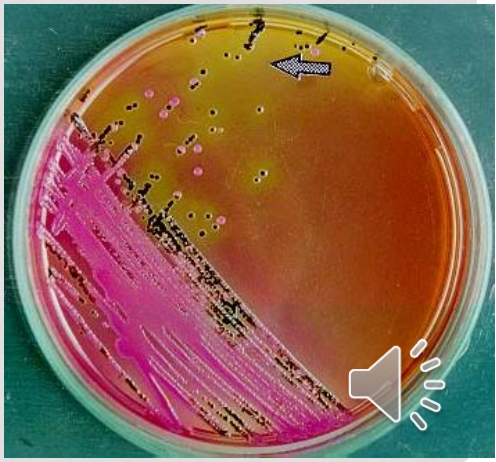


MODERATELY/HIGHLY SELECTIVE MEDIUM

- **Salmonella-Shigella (SS) agar**
  - Bile salts (high concentration), sodium citrate, brilliant green dye : inhibit gram-positive bacteria, many gram-negative bacteria including coliforms
  - Neutral red : indicator
  - **Lactose: CHO source**
  - **Sodium thiosulfate, ferric citrate : detect H_2S**
  - **Permit growth of**

- ***Salmonella***: H_2S production
  - **colorless colony w/ black center**

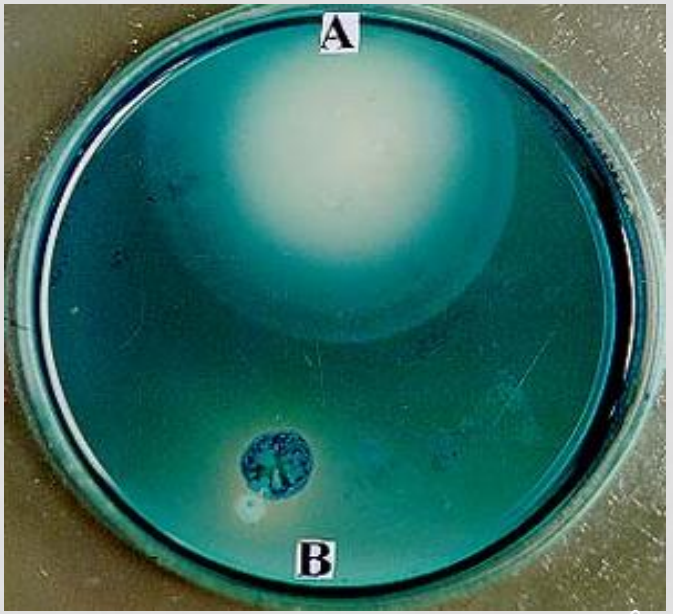
- ***Shigella:*** no H_2S production
  - **coloress colony without black center**



Highly Selective medium

MSRV (Modified Semi-Solid Rappaport Vassiliadis) agar

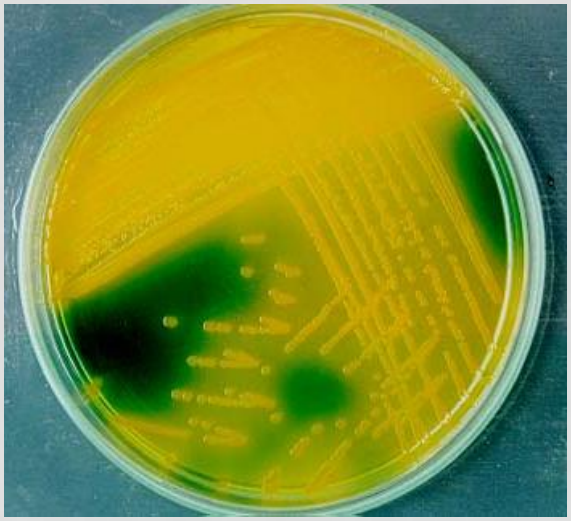
- Detection of motile *Salmonella* at 42^0C
- Highly selective for nontyphoidal *Salmonella*
- Concentration agar 0.27% (motility medium 0.4%)
- pH 5.2† (5.1 to 5.4) @ 25˚C
- Novobiocin, low pH, high $MgCl_2$ concentration and malachite green inhibit the growth of Gram-positive flora



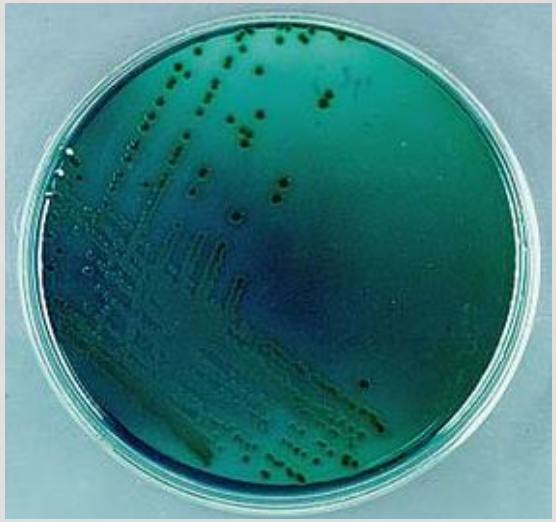
THIOSULFATE CITRATE BILE SALT SUCROSE (TCBS) AGAR

- Oxgall: inhibit Gram-positive bacteria
- Alkaline pH: pH 8.8 ± 0.2
  - inhibit most bacteria, but enhance growth of *Vibrio* spp.
- Sucrose : differential medium
- Bromthymol blue: indicator
- ***Vibrio* spp.**

V. cholerae



V. parahaemolyticus



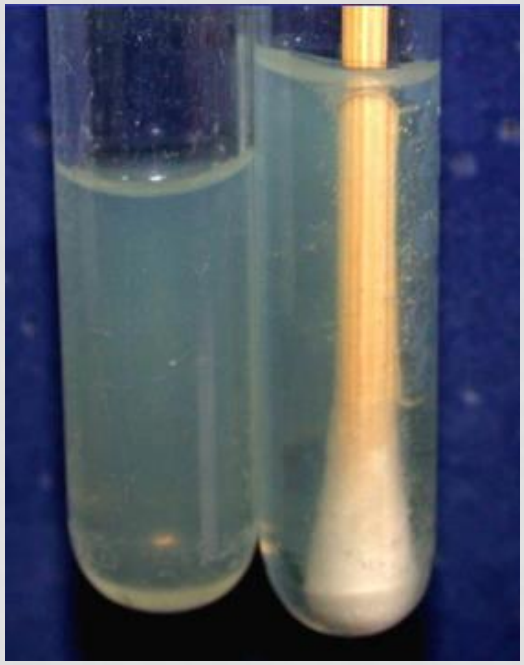
Enrichment broth for *Salmonella/Shigella*



tetrathionate broth

- bile salt inhibit : gram positive organism
- tetrathionate inhibit :normal intestinal flora
- brilliant green inhibit :gram negative and gram positive
- Select for *Salmonella* and *Shigella* spp.(8-12 h)
- If incubation too long normal flora will overgrowth

Enrichment broth for *Vibrio*

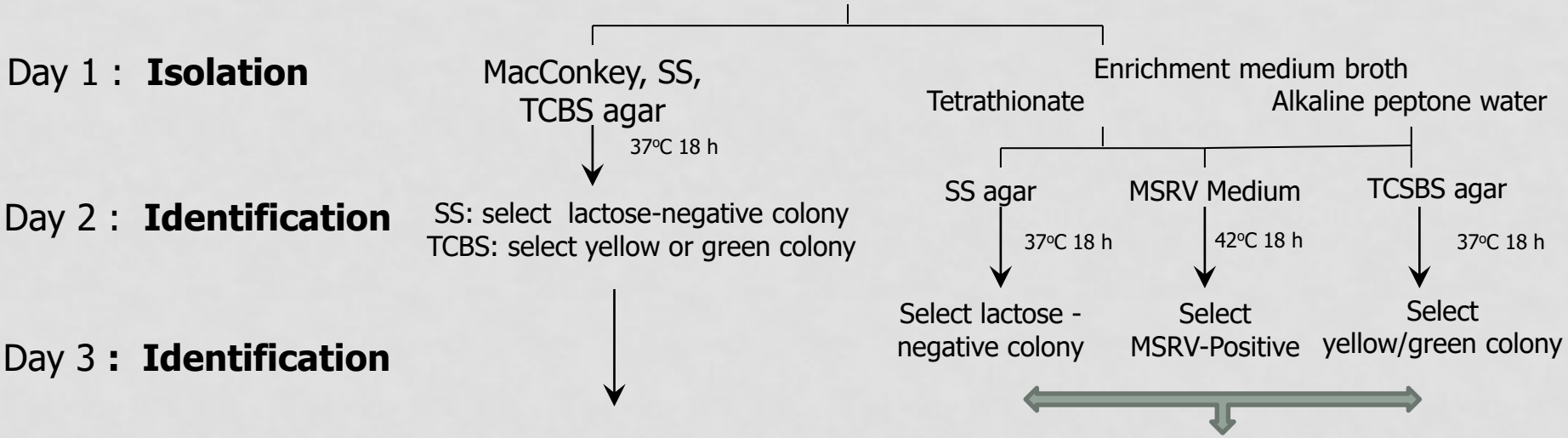


Alkaline peptone water

- Peptone
- NaCl (2% w/v)
- **Alkaline pH: 8.6±0.2 at 37°C**
  - **Inhibit most flora**

Stool Culture for Enteric Pathogens

Specimen in sterile container or Cary-Blair transport medium



Day 4-5 : **Report**

- (*Salmonella* group…..) was isolated.
- Negative for enteric pathogens.

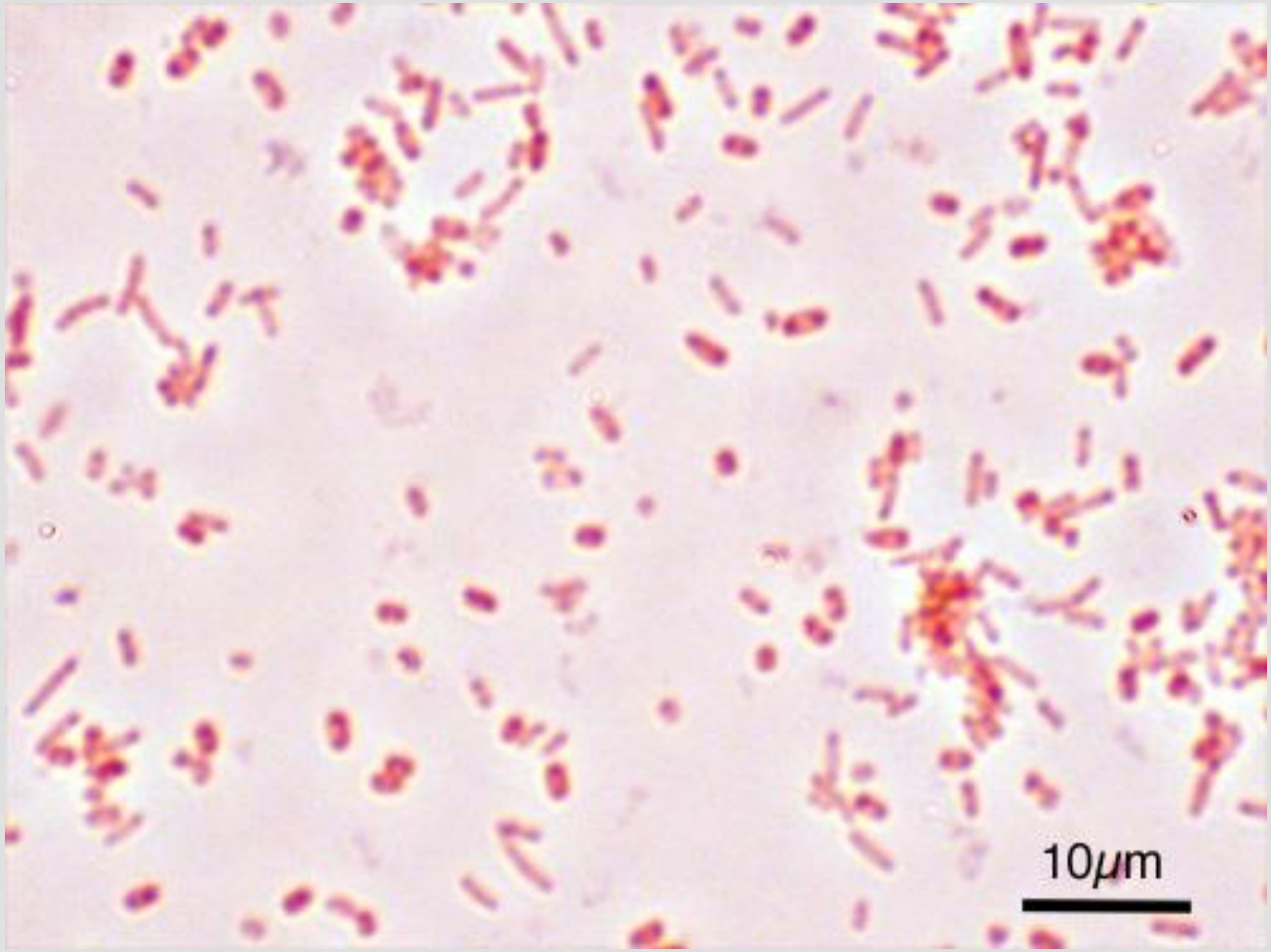
(No *Salmonella, Shigella, Vibrio, Aeromonas, Plesiomonas* were isolated.)

Laboratory activities

2. Gram staining of

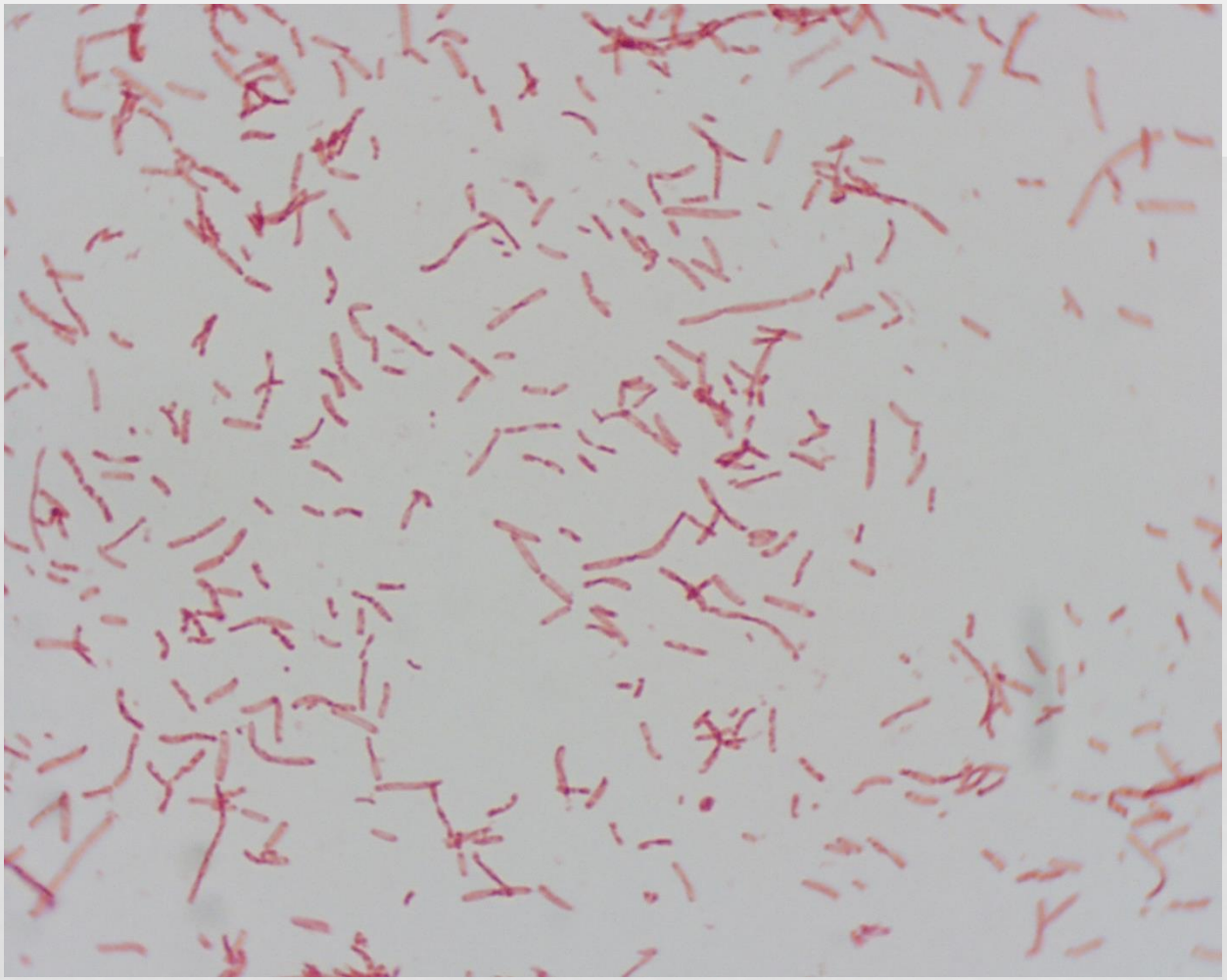
- *Salmonella, Shigella, Vibrio* spp. *Bacillus cereus, Clostridioides/Clostridium difficile*
- *Campylobacter jejuni*
  - Counter stain with 1:20 Carbol fuchsin

SALMONELLA



https://commons.wikimedia.org/wiki/File:Salmonella_Typhimurium_Gram.jpg

SHIGELLA

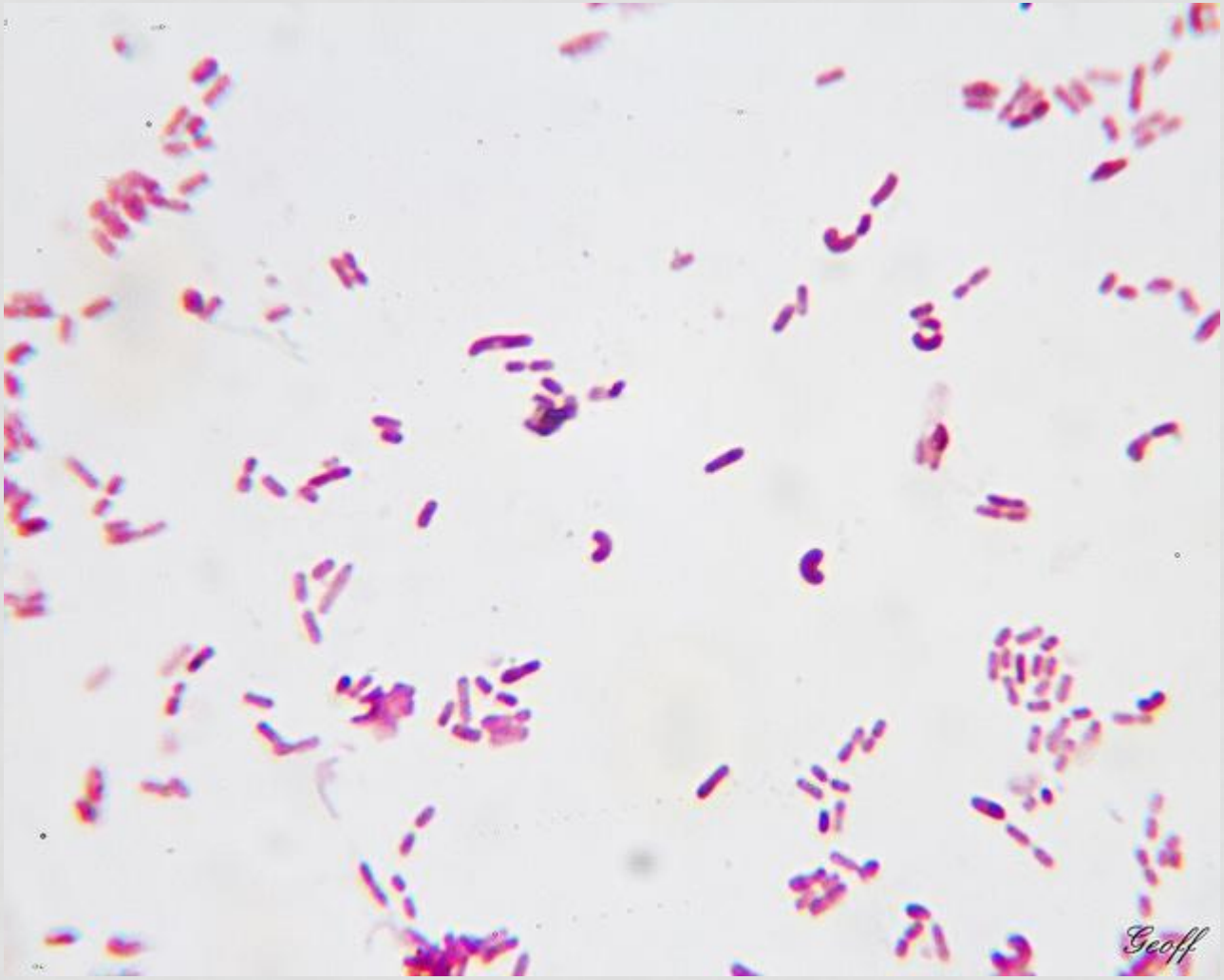


Gram-stain of Shigella sonnei from the previous image (magnification x100). Courtesy of Rita Yee, MT (ASCP) SM

Date of download: 11/28/2016

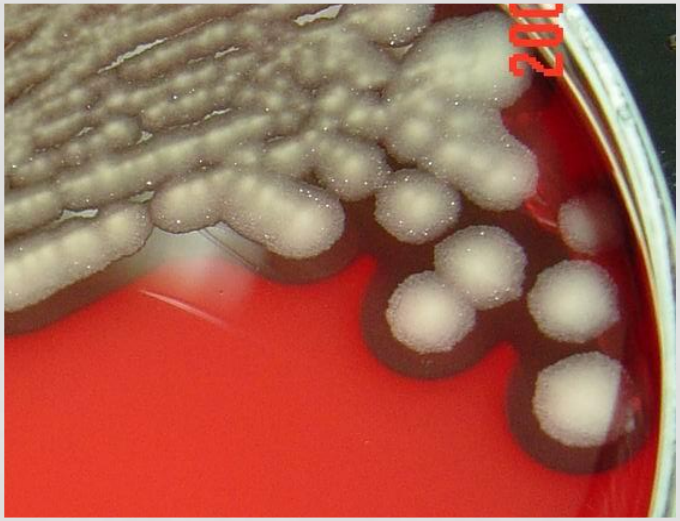
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VIBRIO SPP.



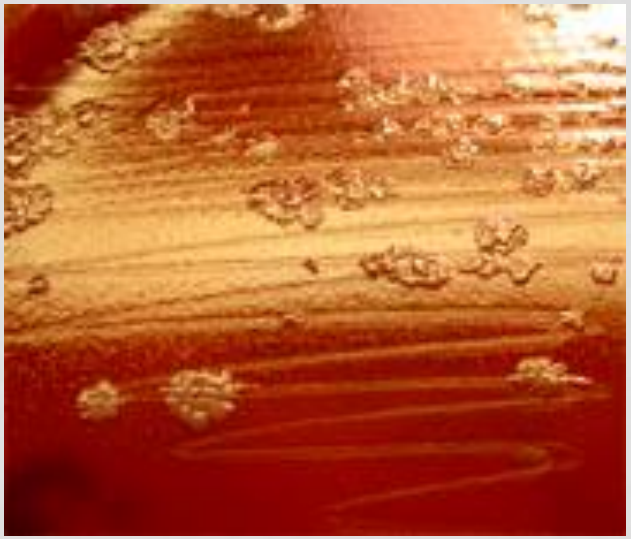
http://lacasamorett.com/foxgallery/vibrio-cholerae-gram-stain.html

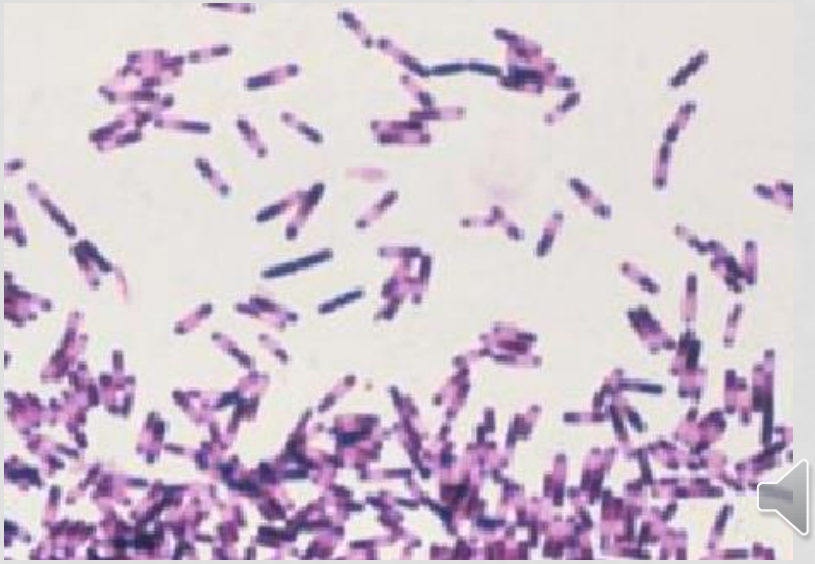
B.cereus





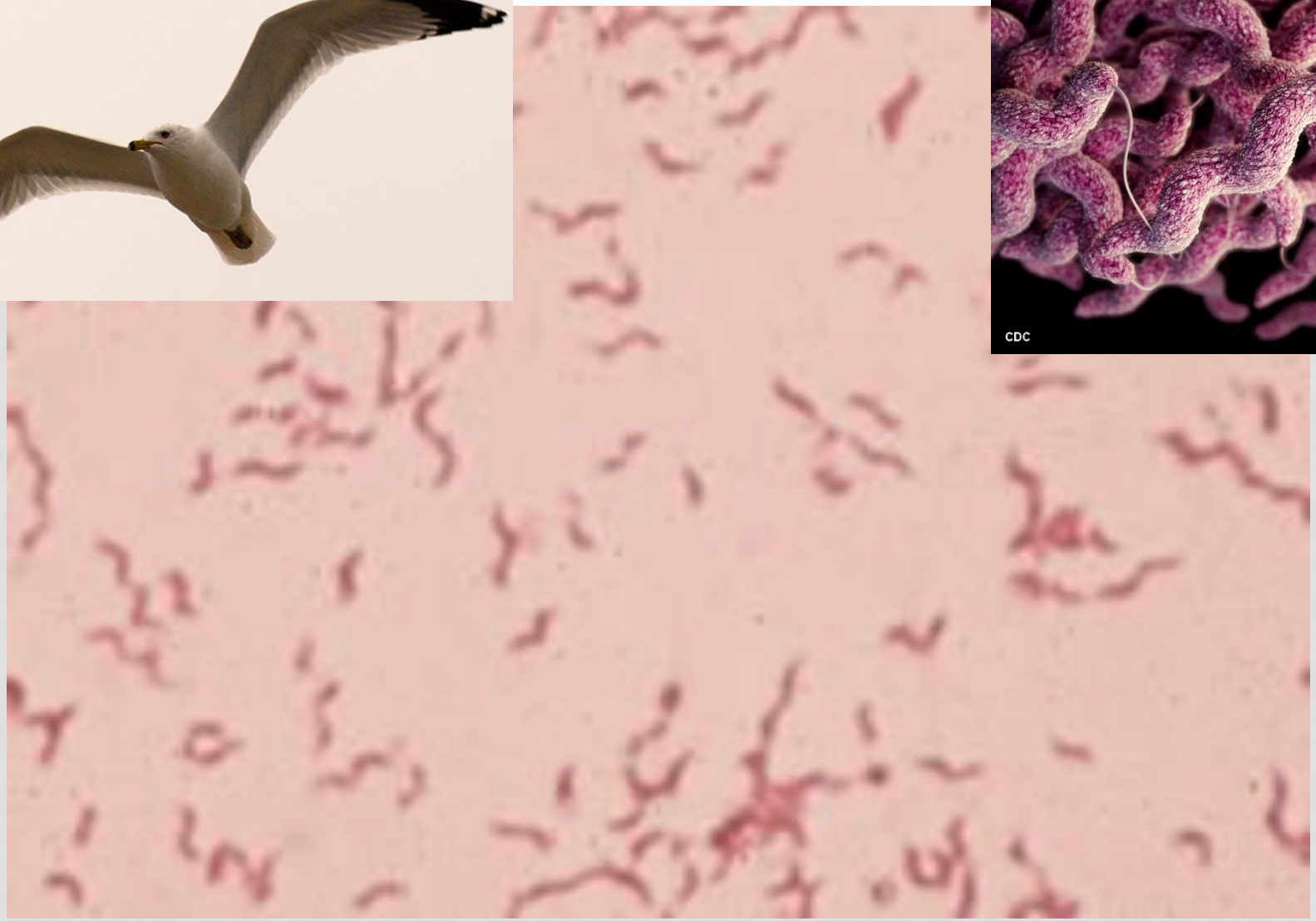
C. difficile





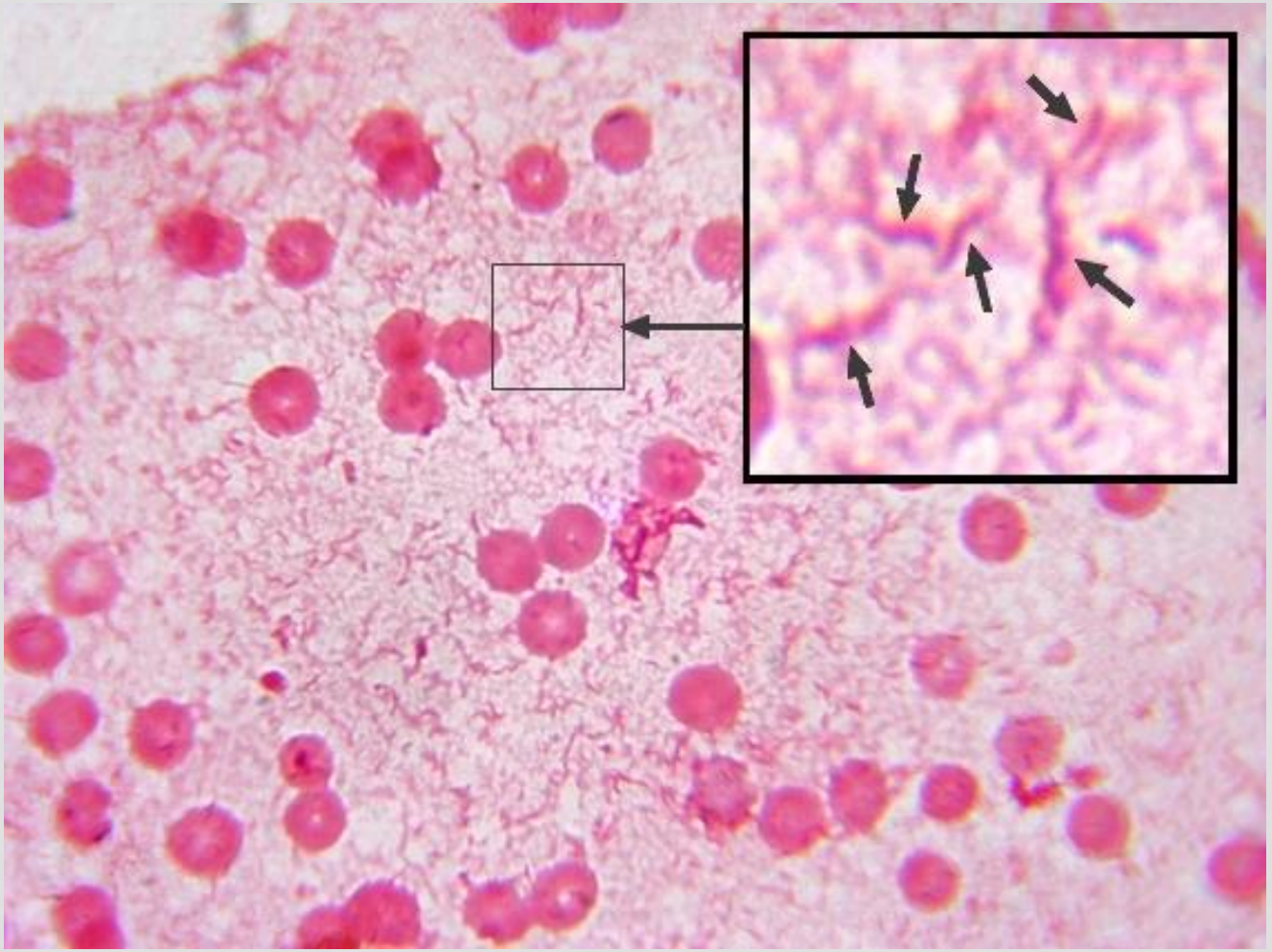
CAMPYLOBACTER JEJUNI







CAMPYLOBACTER JUJUNI IN STOOL GRAM STAIN



http://thunderhouse4-yuri.blogspot.com/2013/02/helicobacter-heilmanii.html

3. Test for the presence of toxigenic *Clostridioides/Clostridium difficile* in stool samples

Example: membrane enzyme immunoassay

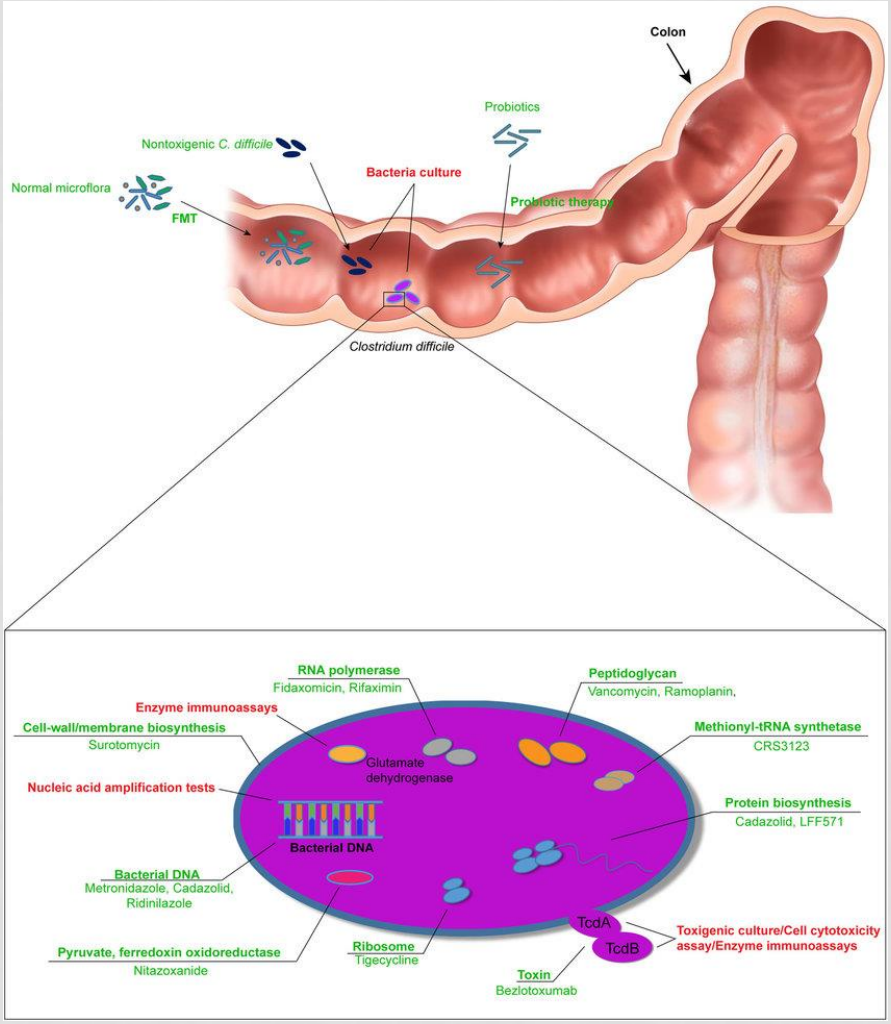
C. DIFF QUIK CHEK COMPLETE® kit

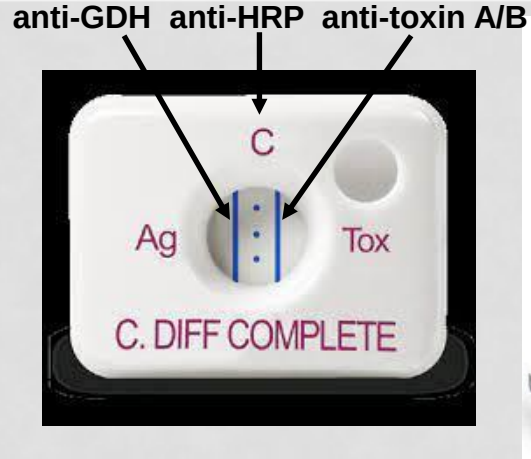
- Use fecal specimens within 72 h of collection
- Glutamate dehydrogenase (GDH): positive for either toxigenic and non-toxigenic *C. difficile*
- Toxin A+B+ or A-B+

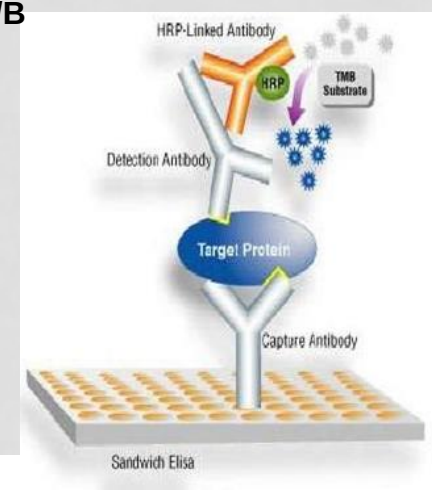
* A+B- strain: very rare

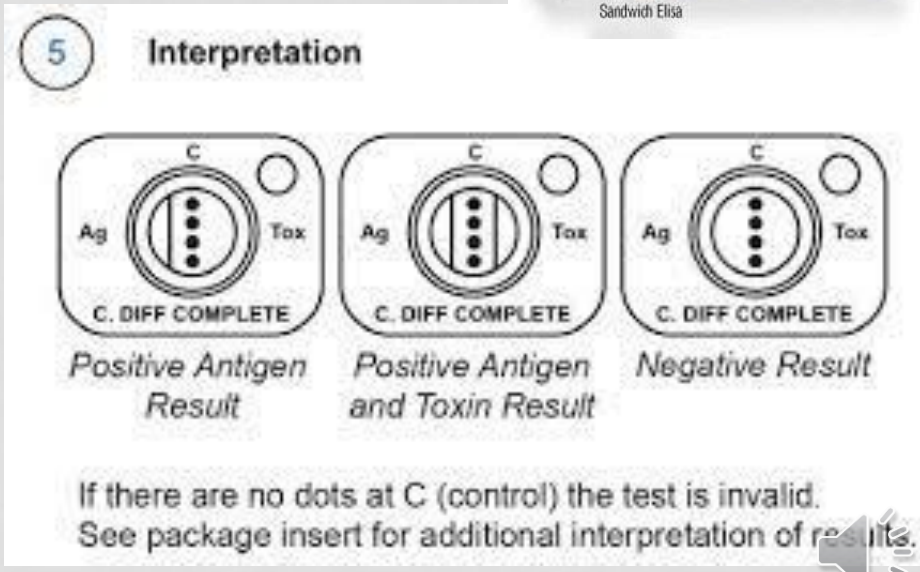


DETECTION OF TOXIGENIC *CLOSTRIDIOIDES/CLOSTRIDIUM DIFFICILE*









Emerg Microbes Infect. 2018; 7: 15.

Summary of Laboratory Activities

1. Stool culture: differential and selective media

- MacConkey agar: *Salmonella* spp.
- SS agar: *Salmonella* spp., *Shigella* spp.
- MSRV agar: *Salmonella* spp.
- TCBS agar: *Vibrio* spp

2. Gram staining

- *Salmonella, Shigella, Vibrio* spp. *Bacillus cereus, Clostridioides/Clostridium difficile, Campylobacter jejuni*

3. Detection of toxigenic *C. difficile* in stool samples

- Toxin A/B

